Bring Your Own Assurance: Composable Test and Evaluation for Agentic Computer Vision with CheckMAITE

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Abstract

AI systems are being applied to more stages of model development, including preparing data, running experiments and creating candidate models. As AI capabilities improve and these workflows become more autonomous, they create a growing need to evaluate candidate models and datasets consistently. We present CheckMAITE, a composable computer-vision test-and-evaluation (T&E) framework that separates candidate development from evaluation. MIT Lincoln Laboratory's MAITE protocols provide the common interface that makes this composition possible. Using these protocols, CheckMAITE runs built-in or organization-specific assurance tests. The resulting evidence can guide an AI agent's next step or support human or external review. We evaluate the design with working demonstrations and an assurance profile for a SeaDronesSee object detector. Together, these capabilities provide a practical foundation for incorporating repeatable assurance checks, transparent evidence, and accountable oversight into increasingly automated AI development.

Candidate Artifacts Need a T&E Boundary

AI systems increasingly generate and revise candidate artifacts during model development, from research assistants to evaluator-guided program search (Lu et al. 2024; Schmidgall et al. 2025; Romera-Paredes et al. 2024; Jiang et al. 2025; Novikov et al. 2025). As candidate production becomes more automated, evaluation needs a stable boundary independent of how a candidate was produced. In computer vision, assurance may require task performance, data quality, robustness, explanations, and review artifacts rather than a single score. CheckMAITE addresses this need with a composable evaluation harness that brings organization-selected evidence together and returns feedback for subsequent work or review.

We use *candidate artifact* for a model or dataset presented for evaluation, regardless of how it was produced. CheckMAITE neither creates nor manages that artifact. Our research question is: *Can one computer-vision T&E harness provide artifact-neutral, extensible, and reusable assurance feedback that autonomous clients can act on?* We assess heterogeneous capability execution, machine- and human-readable evidence, extension mechanisms, and agent-facing feedback.

CheckMAITE is an open-source Python framework for computer-vision T&E. It was developed as an integration point for tools in the Joint AI Test Infrastructure Capability (JATIC) program (OpenTeams and CheckMAITE Contributors 2026). CheckMAITE builds its capability, execution, and evidence layers on MAITE's common interfaces. Our contribution is to characterize and exercise that composition as a bring-your-own-assurance framework for agentic computer vision: candidate production remains separate, while structured evaluator feedback can guide the agent's next action or be escalated for review.

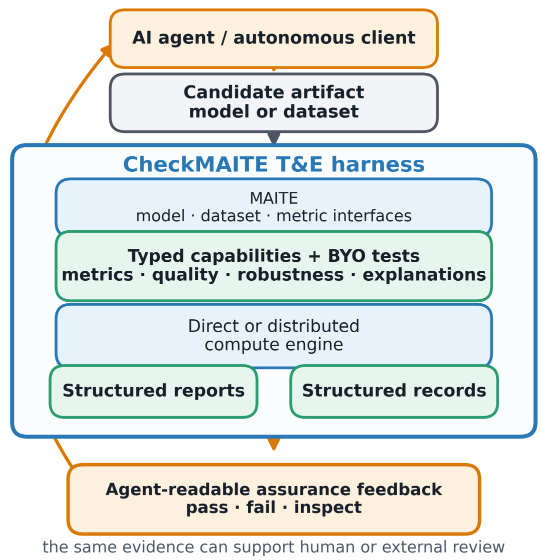

Figure 1: MAITE is the common artifact interface. CheckMAITE returns structured assurance feedback to the agent; the same evidence can support human or external review.

A Composable Assurance Harness

Figure 1 follows an agentic feedback cycle from candidate artifact to assurance evidence and back to the client. The harness combines component contracts, test capabilities, execution backends, reusable computation, reports, and structured records.

MAITE as the composability interface. MIT Lincoln Laboratory's MAITE defines structural protocols for models, datasets, metrics, dataloaders, and augmentations (MIT Lincoln Laboratory and MAITE Contributors 2026). These are the composability interface consumed by CheckMAITE capabilities.

Composable assurance tests. CheckMAITE introduces a *Capability* abstraction for packaging the configuration, execution, and outputs of a computer-vision assurance test. Its built-in capabilities include the standard model-evaluation process through MAITE, data-quality analyses from DataEval (ARIA and DataEval Contributors 2026), and Kitware's NRTK tests for evaluating model robustness to perturbations (Hu et al. 2024). Organizations can add their own assurance capabilities through versioned plug-ins, so new tests can join the same execution and evidence workflow. CheckMAITE reports compatibility or loading problems, while the organization remains responsible for the trustworthiness and scientific validity of contributed tests.

Evidence for people and machines. CheckMAITE records what assurance test was run, how it was configured, and which candidate artifacts were evaluated. Tests can produce structured reports for interpretation and structured records for comparison, automation, and later review. These outputs give AI agents concise feedback while preserving richer evidence for humans and external systems.

Agent Feedback and Accountability

CheckMAITE runs assurance tests and returns structured evidence. Test authors may define criteria and feedback such as pass, fail, or inspect, which an agent or orchestrator can use to continue, revise, request another test, or escalate for review. These signals express the meaning assigned by the test; they are not universal safety judgments. The surrounding system remains responsible for linking results to the correct artifacts, controlling access, and setting deployment policy. Here, *auditable* means that an evaluation and its evidence can be traced and reviewed later.

Agent-facing workflow. An AI client submits a candidate artifact and requests an assurance test through a common interface. CheckMAITE returns feedback that the client can use to revise the candidate, request further evaluation, or ask for review. This allows agents to use several assurance tools through one consistent workflow, while the same evidence remains available to people and external systems.

Evaluation

The companion artifact evaluates whether CheckMAITE can support one assurance workflow across different candidate artifacts, tests, evidence products, and execution settings. Most integration checks use synthetic artifacts. A separate profile applies the workflow to a real SeaDronesSee candidate model and a development subset.

Agent-facing feedback. A deterministic client submits a candidate artifact and requests an assurance test. CheckMAITE rejects a request outside the permitted set and returns structured evidence for a valid request. Data-quality and robustness tests use the same MAITE-based interface and produce reports and records that can be consumed by an agent, a person, or another system. This demonstrates the feedback workflow rather than adaptive agent behavior.

Bring-your-own assurance. The artifact installs a small organization-defined assurance test through CheckMAITE's plug-in interface. CheckMAITE accepts the compatible plug-in, rejects an incompatible version, and returns the test's result and supporting evidence through the same workflow as built-in tests. This shows how organizations can add their own criteria without changing the harness. The example test is intentionally simple and is not presented as a validated assurance method.

SeaDronesSee assurance profile. We applied CheckMAITE to a YOLO11n-P2 detector (Ultralytics 2026) on a deterministic 78-image development subset of SeaDronesSee Object Detection v2 (Kiefer et al. 2023) covering all five classes. The profile combined standard model evaluation and an NRTK brightness sweep, then applied an organization-defined minimum class floor to CheckMAITE's per-class result. Aggregate mAP@50 was 0.853 and passed an illustrative 0.80 threshold. Performance retention across brightness factors 0.75–1.25 was 0.984 and passed a 0.95 threshold. However, AP@50 for the life-saving-appliance class was 0.550, below the 0.60 class floor. The profile therefore returned `inspect` and recommended investigating that class result and the development-set coverage before deciding whether to revise the candidate and repeat the tests.

This use case shows why composable assurance is more useful than a single aggregate score: an agent receives both the successful checks and a specific next action. The subset is a development sample, and the thresholds are illustrative rather than preregistered; the profile does not constitute deployment certification. The checkpoint, images, labels, and sample IDs are not distributed, so the package retains aggregate results, CheckMAITE run identifiers, and provenance hashes and cannot rerun this profile without separately obtained assets.

Operational Benefits and Boundaries

CheckMAITE brings assurance tests, execution, and evidence into a shared workflow. Table 1 summarizes its main operational benefits and the responsibilities that remain outside the harness.

Feedback, not automatic decisions. CheckMAITE makes assurance results visible and structured, but it does not decide what they mean for deployment. Agents and reviewers can use the evidence to choose a next step, while test criteria and policy remain externally defined.

Execution and reuse. Assurance tests can run in local or distributed settings, and previous evidence or computation can be reused when appropriate. Their reliability still depends on the surrounding infrastructure and an accurate understanding of the candidate being evaluated.

Property	Benefit	Boundary
Composable tests	Combine and extend assurance checks	A new test still requires validation
Shared interfaces	Evaluate different candidate artifacts consistently	Integrations must represent artifacts correctly
Structured feedback	Support agent actions and human review	Feedback is not a universal safety judgment
Reusable evidence	Compare and revisit evaluations	Reuse depends on reliable artifact identity
Flexible execution	Use local or distributed resources	Scale and security depend on deployment

Table 1: General operational benefits and boundaries.

Evidence for different audiences. Reports support interpretation, while structured records support automation and comparison. Keeping both helps agents, reviewers, and external systems work from the same evidence.

Assurance Context and Related Work

Agent benchmarks and program-search systems show that increasingly autonomous AI workflows depend on evaluator feedback (Huang et al. 2024; Chan et al. 2024; Romera-Paredes et al. 2024; Jiang et al. 2025; Novikov et al. 2025). For safety and assurance, however, feedback must do more than optimize one score: it must combine relevant evidence, preserve context, and remain available for accountable review.

Existing work addresses different parts of this need. In 2023, the Chief Digital and Artificial Intelligence Office (CDAO) described its establishment of JATIC as its "keystone T&E effort" and as an interoperable set of software capabilities for AI algorithm T&E. It said JATIC would test model robustness, resiliency to adversarial attack, humans' ability to understand and trust model outputs, and competence, while easing deployment, integration, and compatibility with the AI/ML pipelines adopted by different Department of Defense organizations (Martell 2023). Developed under JATIC, MIT Lincoln Laboratory's MAITE supplies common types and structural protocols intended to enable interoperability and composition across T&E tools (MIT Lincoln Laboratory and MAITE Contributors 2026). DataEval supplies dataset analyses, while NRTK supplies natural-perturbation robustness tooling (ARIA and DataEval Contributors 2026; Hu et al. 2024). CheckMAITE provides the shared computer-vision T&E workflow that brings such capabilities together. In our architecture, organizations can compose assurance tests, apply them consistently to candidate artifacts, and return structured evidence to agents and reviewers.

Conclusion

Increasingly automated AI development creates an assurance gap: candidate models and datasets can change faster than evaluation practices can be assembled and reviewed. CheckMAITE addresses this gap by using MIT Lincoln Laboratory's MAITE protocols to let organizations compose standard and bring-your-own computer-vision assurance tests and return structured evidence to AI agents and accountable reviewers. Assurance cannot rely on a single aggregate score. Capability-defined feedback may guide revision or escalation, but it does not authorize deployment. By turning disconnected checks into a shared, reviewable evidence workflow, CheckMAITE provides a practical foundation for keeping increasingly automated computer-vision development testable, reviewable, and accountable.

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